Nilesh Jain

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SUMMARY

Organized and self-driven **Full-Stack Developer** with **1.5 Years** of industry experience in building scalable web and mobile applications, firmware testing, and **Embedded Systems**. Proficient in **Python**, **React**, **Embedded C/C++**, **Android Studio**, and **RESTful API** development. Adept at **Agile** methodologies, **test automation**, and **cross-platform** development using **MERN**, **Flutter**, and **Docker**. **Passionate** about continuous learning and applying new technologies to deliver robust, user-centered solutions.

EDUCATION

University of Illinois at Chicago • BS Computer Science | Minor in Mathematics **2023**
Dean's List | Honors College (Research)

WORK EXPERIENCE

Software Engineer (Embedded/Firmware Development & Testing) – SPG America LLC, New Jersey **Dec 2023 – July 2024**

- Engineered test development using **C**, **C++**, **CPPUNIT** architecture and **python** using **Pytest** for visualization and followed **test plans** through automation, collecting and analyzing test results to test multiple devices in the field to get necessary logs.
- Helped clients with IT Solutions, **Troubleshooting** and **Debugging** which led to company expansion and client trust.
- Used **Jenkins** as build management tool for continuous integration process along with **Git** as **Version Control** tool.
- Developed and deployed **Bluetooth/Ethernet**-enabled applications; **enhanced firmware stability** by **30%** using **PuTTY**-based serial debugging.
- Created and maintained **Python modules**, made sure it was easy to deploy and leveraged **PYPi** for distribution and management of packages and was involved in bug fixing of modules that were raised by the testing teams in the application during the **integration testing** phase.

Software Engineer (Front-End Developer) – SPG America LLC, New Jersey **Jan 2023 – Nov 2023**

- Developed and maintained mobile applications using **Android Studio** and **Kotlin**, and integrated **APIs** and third-party services like **Twilio**, **SendGrid** to include email and messaging functionalities, resulting in **18% increase** in **user engagement**.
- Leveraged the skills in **React**, **Node.js**, **Express.js** and **TypeScript** to assist in the development of a data-driven web application and **RESTful APIs**. This project improved **client-side efficiency** by **80%** and was completed in 4 months.
- Implemented user intuitive mobile/web designs for user enrollment using **Android Studio**, **React**, **Tailwind** for UI/UX, **Rest APIs**, and extensive **SQL** querying for decreasing client-side processing for a **streamlined experience**.
- Wrote **unit tests** with **96% code coverage** for all mobile/web applications using **Jest** and **Junit,** and **integration tests** for APIs to enable test automation, ensuring high product quality.
- Attended weekly/daily Standups, Retrospective, Sprint Planning, Code Review and completed story points on **JIRA** tool and worked in **Agile Methodology** in a 2-week sprint environment.

PROJECTS

Vehicle Rental App - GoCarz (Android Studio, Kotlin, Java, Dart, Flutter, Firebase)

- Built a feature-rich mobile app using **Android Studio** and **Firebase** for renting cars easily from home all over **India**.
- Designed a user-friendly UI/UX and implemented **multi-activity flow** using **Parcelable** model classes in **Java** and **Kotlin** for backend operations, ensuring seamless integration with native Android components.
- Implemented **admin-side** features for shop owners to add vehicles, set prices and add quantities and **client-side** functionality to fetch and select vehicles added to book, create profiles and rent vehicles for a certain period of time with updated history using **Firebase Realtime Database**.
- Utilized **Gradle** for efficient project management and added **authentication** for Login and Sign-Up (**google-services**).

American Sign Language to Text and Speech Converter (Python, AI/ML, Tensorflow, Mediapipe, Computer Vision)

- Enabled **real-time** hand gesture-to-text translation, improving accessibility for hearing-impaired users.
- Leveraged webcam input, **Mediapipe** for hand landmark detection, and a **Convolutional Neural Network (CNN)** model via **TensorFlow** to recognize and classify hand signs and integrated **CVZone** for visual overlays on the live video feed.
- Achieved high classification accuracy through **model training** on gesture datasets; supported all 26 English alphabets.

Exercise Break App (React.js, Android Studio, Express, Node.js, MongoDB, Kotlin, HTML, CSS)

- Designed and implemented the “Exercise Break App”, an advanced wearable mobile application for real-time heartbeat and location tracker using **COTS** software, **APIs** (Google Maps) and **MySQL** for relational database management.
- Leveraged **Android Studio**, **Kotlin** for mobile application with interactive widgets and **React**, HTML, CSS for user-friendly UI web application.
- Implemented **IoT connections** between wearables and mobile applications using **Bluetooth** protocols to provide users with accurate and optimized breaks.

SKILLS

Languages: Python | Java | C/C++ | Kotlin | SQL | JavaScript | R
Front-End: React | HTML/CSS | Tailwind
Back-End: Node.js | Express.js | MongoDB | Firebase
Mobile: Android Studio | Flutter | Dart
Tools: Git | GitHub | Gradle | Docker | Jenkins
Testing: Pytest | JUnit | Jest | CPPUnit
Protocols: Ethernet | Bluetooth